

# Milestone 4

## Proof of Achievement Report

Project Catalyst #1200045: MPC as a Layer 2 Service to Cardano

Profila AG x Partisia Blockchain

Date: 2026-03-06

Network: Cardano Preprod + Partisia Blockchain Testnet

### 1. Executive Summary

Milestone 4 completes the end-to-end MPC pipeline. The MPC computation result is displayed to the initiator, the full cross-chain flow from Cardano query to Partisia MPC result is demonstrated, and the final closeout report and video are publicly available.

**All acceptance criteria are met. Project is complete.**

### 2. Acceptance Criteria - Evidence Matrix

| # | Criterion                         | Status | Evidence                         |
|---|-----------------------------------|--------|----------------------------------|
| 1 | MPC result displayed to initiator | PASS   | Section 3 - result=5 on-chain    |
| 2 | PDF documentation in English      | PASS   | This document                    |
| 3 | Process of displaying MPC result  | PASS   | Section 4 - show_result.ts flow  |
| 4 | YouTube video of process          | PASS   | youtu.be/2YYvELDzrGo             |
| 5 | E2E demo on Cardano + PBC testnet | PASS   | Section 5 - full pipeline        |
| 6 | Final closeout report public      | PASS   | Section 7 - closeout-report.json |
| 7 | Final closeout video public       | PASS   | youtu.be/jwwfdTSUS2w             |

### 3. MPC Query Result

#### 3.1 Result Displayed to Initiator

The MPC computation result is fetched from the live PBC contract state via REST API and displayed to the initiator via terminal output and HTML evidence page.

| Property             | Value                                              |
|----------------------|----------------------------------------------------|
| Query Type           | age_threshold (count users with age > 18)          |
| Dataset              | profila_test_v1                                    |
| Result               | 5 (5 out of 5 submitted users have age > 18)       |
| Participants         | 50 (dataset), 5 (submitted secrets)                |
| Computation Method   | MPC across 4 PBC ZK nodes                          |
| Result Source        | on-chain (live PBC contract state)                 |
| Computation Complete | true                                               |
| Privacy              | Aggregate result only - no individual data exposed |

### 3.2 Result Evidence Files

MPC result JSON: relay/mpc-result.json

HTML result display: relay/result-display.html

Result display script: relay/show\_result.ts

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## 4. Process of Displaying the MPC Result

### 4.1 How the Result Is Retrieved and Displayed

1. show\_result.ts calls PBC REST API: GET /chain/contracts/{address}
2. Receives serializedContract (base64-encoded binary state)
3. Decodes ProfilaMpcState: administrator, dataset\_id, query\_type, min\_participants, result, computation\_complete
4. Extracts the aggregate result value (i64)
5. Displays result in terminal with formatted output
6. Generates HTML evidence page with cross-chain links
7. Writes mpc-result.json with full provenance chain

### 4.2 Computation Trigger

The MPC computation was triggered via the PBC testnet browser with 100M gas:

| Property        | Value                               |
|-----------------|-------------------------------------|
| Action          | start_computation (shortname 0x01)  |
| Trigger TX      | d836a6fc075e9ce7...f07c54cc         |
| Gas Fee         | 100,000,000 (100M)                  |
| 4 MPC Nodes     | All committed results successfully  |
| Contract Result | computation_complete=true, result=5 |

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## 5. End-to-End Process - Cardano to MPC Result

The complete pipeline from query initiation on Cardano to MPC result:

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Stage 1: Cardano SC Deployment  -> Aiken MpcRequest UTxO on preprod
Stage 2: Relay Detection        -> relay.ts polls Blockfrost, detects datum
Stage 3: PBC Contract Deployment -> ZK contract on PBC testnet
Stage 4: Secret Submission      -> 5 real ZK transactions via PBC SDK
Stage 5: MPC Computation        -> Triggered via browser, 4 nodes compute
Stage 6: Result Display         -> show_result.ts decodes on-chain state

```

### 5.1 Cross-Chain Verification Links

Cardano TX (query initiation):

<https://preprod.cardanoscan.io/transaction/50d28fd2bd263a84485b45280afd19bb4c3e20c24e9a4b52b6eac20418d53a57>

Cardano Script Address:

[https://preprod.cardanoscan.io/address/addr\\_test1wzut662xhd8e4jq97fpdsml2pyqerejwzfg24red2n7gceczhkwaz](https://preprod.cardanoscan.io/address/addr_test1wzut662xhd8e4jq97fpdsml2pyqerejwzfg24red2n7gceczhkwaz)

PBC Contract (result + state):

<https://browser.testnet.partisiablockchain.com/contracts/039ed9214602f2a93eb05411f852fa78a476607634>

## 5.2 E2E Run Log

Full 6-stage execution log with all TX hashes: docs/evidence/e2e-run-log.json

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## 6. Test Summary

| Suite           | Tests | Passed | Framework   |
|-----------------|-------|--------|-------------|
| Aiken (Cardano) | 6     | 6      | aiken check |
| PBC (Rust)      | 11    | 11     | cargo test  |
| Total           | 17    | 17     | All passing |

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## 7. Final Closeout

### 7.1 Closeout Report

Publicly available in the repository:

[https://github.com/Profila/Aiken-Partisia\\_MPC/blob/main/docs/evidence/closeout-report.json](https://github.com/Profila/Aiken-Partisia_MPC/blob/main/docs/evidence/closeout-report.json)

### 7.2 Closeout Video

M4 Result & Closeout Demo: <https://youtu.be/2YYvELDzrGo>

Combined M1-M4 Full Demo: <https://youtu.be/jwwfdTSUS2w>

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## 8. Repository

GitHub: [https://github.com/Profila/Aiken-Partisia\\_MPC](https://github.com/Profila/Aiken-Partisia_MPC)

Result display: relay/show\_result.ts

MPC result: relay/mpc-result.json

Computation trigger: relay/trigger\_computation.ts

E2E run log: docs/evidence/e2e-run-log.json

Closeout report: docs/evidence/closeout-report.json

Evidence page: docs/poc-evidence.html

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## 9. Video Demos

M4 Result & Closeout: <https://youtu.be/2YYvELDzrGo>

Combined M1-M4 End-to-End: <https://youtu.be/jwwfdTSUS2w>

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